

Graham C. McLaughlin

grahamcm@email.sc.edu | (704) 999-7628

Education

University of South Carolina, Columbia, SC

PhD., Evolutionary Genomics

BS, Biological Sciences, *Magna Cum Laude*

August 2022 – May 2026

August 2018 – May 2022

Research Interests:

Population Genomics, Phylogenomics, Biogeography

Peer-Reviewed Publications

Kincaid, S., **McLaughlin, G. C.**, Sudak, S., & Hollis, B. (2025). Seminal Fluid Protein Acp29AB Shifts Egg-Laying Timing in *Drosophila* Without Detectable Effects on Female Fitness. *Ecology and Evolution*, 15(12). <https://doi.org/10.1002/ece3.72677>

Publications in Prep

McLaughlin, G.C., Baquero, V., Pozo, M.J., de Lourdes Torres M., & Hollis B (2026). Colonization History of the Galápagos Carpenter Bee *Xylocopa darwini* Revealed Through Whole Genomes. To be submitted to *Molecular Phylogenetics and Evolution*. Preprint avail. soon.

McLaughlin, G.C., Baquero, V., Sudak S., & Hollis, B. (2026). 135 Generations of Experimental Reinforcement Strengthens Reproductive Isolation in a Regime-specific Manner and Leaves Predictable Signals of Genomic Divergence. To be submitted to *Evolution Letters*.

Awards and Grants

First Author

- UofSC Department of Biological Sciences Taber Travel Fellowship 2026 (\$1,350)
- UofSC Department of Biological Sciences Taber Travel Fellowship 2024 (\$1,260)
- UofSC Office of the Vice President for Research *Support to Promote Advancement of Research and Creativity* (SPARC) graduate research award (\$4,960)
- 2023 Southeastern Population Ecology and Evolutionary Genetics Conference Mentorship Program Award (Cost of Registration)
- Society for the Study of Evolution Student/Postdoc Evolution 2023 Travel Award (\$500)
- Graduation with Leadership Distinction in Research
- Discover UofSC Conference: 2nd Place Prize, Biology and Biomedical Sciences Poster Session
- 2022 Southeastern Population Ecology and Evolutionary Genetics Conference NSF Registration Award (\$250)
- UofSC Academic Achievement Scholarship Award (\$2000)
- UofSC Class of 22' Outstanding Senior in Biological Sciences Award (\$500)
- UofSC Magellan Scholar Undergraduate Research Grant (\$2800)

Mentored

Sydney Wilson

- USC Magellan Scholar Undergraduate Research Grant – “Effect of male age on precopulatory fitness in *Drosophila melanogaster*” (unfunded)

Marion Sturkie

- USC Magellan Scholar Undergraduate Research Grant – “Response of male sexual behavior to female social environment” (\$2500)

Presentations

*Name in **bold** when either primary author or mentor.

Wilson S, **McLaughlin G**, Hollis B. “Precopulatory Fitness Effects of Male Age and Sexual Maturity in *Drosophila melanogaster*” (Poster) presented at: Discover USC 2025; 2025 Apr 25; Columbia, SC.

*1st place award in Biology and Environmental Science poster session

Sturkie M, **McLaughlin G**, Hollis B. “Exploring how Social Environment Alters Male Mating Behavior” (Poster) given at: Discover USC 2025; 2025 Apr 25; Columbia, SC

McLaughlin G. “Experimental Speciation via Reinforcement: Results from 120 Generations of Experimental Evolution” (Talk) given at: USC Department of Biological Sciences Ecology and Evolutionary Biology Graduate Student Seminar Series; 2025 Apr 11; Columbia, SC.

McLaughlin G, Baquero V, Pozo M, de Lourdes Torres M, Hollis B. “Population Genomics and Biogeography of the “highly vagile” Galapagos carpenter bee” (Talk) given at: 3rd Joint Congress on Evolutionary Biology; 2024 July 27; Montreal, Quebec, CA.

Brosokas T, **McLaughlin G**, Hollis B. “Exploring population-specific male mating behavior differences in laboratory island populations of *Drosophila melanogaster*” (Poster) presented at: Discover USC 2024; 2024 Apr 19; Columbia, SC.

*1st place award in Biology and Environmental Science poster session

McLaughlin G. “Migration between island populations of the Galapagos carpenter bee” (Talk) given at: USC Department of Biological Sciences Ecology and Evolutionary Biology Graduate Student Seminar Series; 2024 Feb 7; Columbia, SC.

McLaughlin G, Hollis B. “Exploring Laboratory Island Populations as Study Systems for Early-stage Speciation” (Poster) presented at: The Southeastern Population Ecology and Evolutionary Genetics Conference (SEPEEG); 2023 Sept. 22-24; Pembroke, VA.

Sturkie M, **McLaughlin G**, Hollis B. “The Effect of Adult Social Environment on *Drosophila melanogaster* Mating Behavior” (Poster) presented at: The South Carolina Governor’s School for Science and Mathematics Summer Research Symposium; 2023 Jul 14; Columbia, SC.

McLaughlin G, Hollis B. “An Empirical Test of the Reinforcement Model of Speciation in *Drosophila melanogaster*” (Talk) given at: Evolution; 2023 Jun 21-25; Albuquerque, NM.

McNulty S, **McLaughlin G**, Hollis B. “Laboratory Island Populations as Speciation Study Systems” (Poster) presented at: Discover USC 2023; 2023 Apr 21; Columbia, SC.

*Honorable Mention in Biology and Environmental Science poster session

McLaughlin G. “Experimental Speciation via Reinforcement: Results from 70 Generations of Experimental Evolution” (Talk) given at: USC Department of Biological Sciences Ecology and Evolutionary Biology Graduate Student Seminar Series; 2023 Feb 27; Columbia, SC.

Kincaid, S, **McLaughlin G**, Hollis B. “Sexual Conflict and Seminal Fluid Proteins in *Drosophila melanogaster*” (Talk) given at: Evolution; 2022 Jun 24-28; Cleveland, OH.

McLaughlin G, Hollis B. “Experimental Speciation using Laboratory Island Populations of *Drosophila melanogaster*” (Poster) presented at: Discover USC 2022; 2022 Apr 22; Columbia, SC.

*2nd place award in Biology and Environmental Sciences poster session

McLaughlin G, “Experimental Speciation using Laboratory Island Populations of *Drosophila melanogaster*” (Poster) presented at: The Southeastern Population Ecology and Evolutionary Genetics Conference (SEPEEG); 2022 Mar 25-27; Rock Eagle, GA.

McLaughlin G, Hollis B. “Experimental Speciation: Sexual and Gametic Reproductive Isolation”
(Poster) presented at: USC Summer Research Symposium; 2021 Jul 29; Virtual Online.

Workshops

Summer bioinformatics workshop

Hosted by Drs. Richard Flamio and Christina Ramstad at the University of South Carolina Aiken

August 2020 – Dec. 2020

Research Experience

Undergraduate Independent Study

Targeted Expression of Seminal Fluid Proteins Using UAS/GAL4 Genetic System

Mentor: Brian Hollis

Dec. 2020 – May 2022

- Tested the validity of two GAL4 drivers and 4 UAS-RNAi transgenic fly stocks
- Conducted genetic crosses to produce flies with targeted gene knockdown
- Quantified gene knockdown using RNA extraction and RT-qPCR

Magellan Scholar Undergraduate Researcher, Department of Biological Sciences

Investigating the Effects of Specific Seminal Fluid Proteins on the Female Postmating Response in D. melanogaster

May 2021 - Present

- Designed and executed numerous experiments testing male mediated effects on female fitness
- Elucidated the role of seminal fluid proteins in sperm competition

Experimental Speciation using Laboratory Island Population of D. melanogaster

- Assisted in design and execution of experiments assessing reproductive isolation due to reinforcement
- Data analysis using linear, generalized linear, and mixed effects models in R

Professional Experience

South Carolina Department of Natural Resources Freshwater Fisheries Intern

2022 Rock Hill Fisheries Management Intern

Supervisors: Robert Stroud, Preston Chrisman

May 2022 – July 2022

University of South Carolina Teaching Assistant

Biol 101L, Biological Principles I (Fall 22')

BIOL 303L, Fundamentals of Genetics Laboratory (Spring 23', Fall 23', Spring 24', Fall 24', Spring 25')

Professional Society Membership

Society for the Study of Evolution, *member*

American Society of Naturalists, *member*

Ecological Society of America, *member*

Service

Secretary, Graduate Association of Biological Sciences

December 2024

- Designed and disseminated a Graduate Student Experience Survey aimed at gauging Biology Graduate student satisfaction and support
 - Received 35 responses shared with department administration
- Organized the 2025 edition of the annual “Data Dinner Symposium”
 - Enjoyed invited two invited speakers followed by catered food funded by the University of South Carolina Department of Student life.

Thurs, March 27th, 2025

NSF Funded Science Outreach

Mon, April 17th, 2023

- Demonstrated mendelian genetics using fruit flies to local Lexington, SC 8th graders.

- Students set up crosses of flies with mendelian phenotypes and predicted which trait was dominant/recessive and homo-/heterozygous.

Future STEM Heroes

- Helped organize local STEM fair for local middle school students at the University of South Carolina
- Joint effort between the UofSC Graduate Association of Biological Science Students (GABS), and the Pharmacy Graduate Student Association.

Reviewer Discover UofSC

- Reviewed 6 posters in the Experiential Learning poster session
- 53rd Annual South Carolina *Junior Science and Humanities Symposium*
- Judged 6 finalist talks for selection to nationals

Mon, April 14th, 2024
Wed, March 19th, 2025

Saturday, April 22nd, 2023
Saturday, Feb 24th, 2024

Additional Experiences

TriBeta Biological Honor Society, Member

- Weekly meetings, research showcases, community service

January 2020 – August 2022

Genetics Club, Member

- Guest speakers, literature reviews, community service

January 2020 – August 2022

Carolina Judicial Council, Student panelist

- Served as panelist on an appellate review board for Office of Student Conduct and Academic Integrity

August 2018 – Dec. 2020

University Housing, Resident Mentor (RA)

- Employed as a resident mentor (RA) for first year students

August 2019 – May 2020

Applicable Skills

Certifications: PADI Open Water Scuba Diver & SSI Advanced Open Water Scuba Diver

Computational: Microsoft Office Suite, R, Java, Bash

Molecular Biology: RNA/DNA isolation/purification, qPCR analysis, gel electrophoresis

Basic Laboratory Skills: Micropipette, microscopy, fluorescence microscopy, and numerous *Drosophila*-related skills